

SYDNEY TECHNICAL HIGH SCHOOL

Name: _____

Teacher : _____



Mathematics Extension 2

HSC ASSESSMENT TASK 2

TERM 2 - 2008

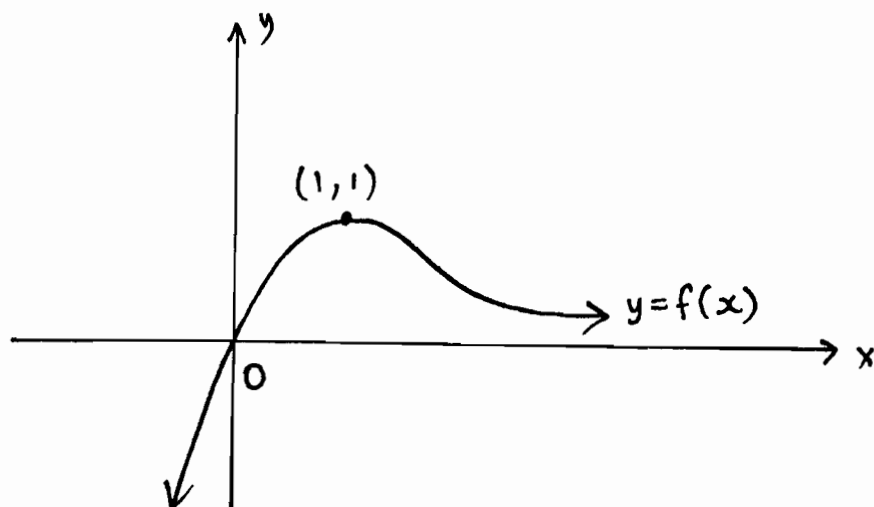
General instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions

QUESTION 1	QUESTION 2	QUESTION 3	TOTAL

QUESTION 1 (18 Marks) Start a new page

a)



Sketch on separate diagrams the graphs of

i $y = f(x+2)$ 1

ii $y = f|x|$ 2

iii $y^2 = f(x)$ 2

iv $y = \ln f(x)$ 2

b) Evaluate

i $\int_1^3 x \ln x \, dx$ 3

ii $\int_0^\pi \sin^3 x \, dx$ 3

c) Find $\int \frac{dx}{\sqrt{5+4x-x^2}}$ 3

d) Evaluate $\int_0^1 \frac{2}{\sqrt{x^2+1}} \, dx$, leaving your answer in exact form 2

QUESTION 2 (16 Marks) Start a new page

- a) i Find the real numbers A, B and C such that

$$\frac{x-2}{(x^2+4)(x+1)} = \frac{Ax+B}{x^2+4} + \frac{C}{x+1} \quad 3$$

- ii Hence, find $\int \frac{x-2}{(x^2+4)(x+1)} dx$ 3

- b) Find the integers m and n such that $(x+1)^2$ is a factor of $x^5 + 2x^2 + mx + n$ 2

- c) If α, β, δ are the roots of the polynomial $4x^3 + 8x^2 - 1 = 0$

- i Find a polynomial with roots α^2, β^2 and δ^2 2

- ii Find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\delta^2}$ 2

- d) Use the substitution $x = 2 \sin \theta$ to find $\int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx$ 4

QUESTION 3 (16 Marks) Start a new page

- a) If $ax^4 + bx^3 + dx + e = 0$ has a triple root, show that $4a^2d + b^3 = 0$ given that a, b, d , and e are non zero integers. 3

- b) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are two variable points on the rectangular hyperbola $xy = c^2$, so that the points P , Q and $S(c\sqrt{2}, c\sqrt{2})$, are always collinear. 8

The tangents to the hyperbola at P and Q intersect at R .

- i Show that the tangent to the hyperbola $xy = c^2$ at the point $T\left(ct, \frac{c}{t}\right)$ has equation $x + t^2y = 2ct$
- ii Show that R has co-ordinates $\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$
- iii Show that $p + q = \sqrt{2}(1 + pq)$
- iv Hence, find the equation of the locus of R

- c) i Find the complex solutions of $z^7 = 1$ 3

- ii Hence, by factorizing over the real numbers, prove that

$$\cos\frac{3\pi}{7} + \cos\frac{\pi}{7} - \cos\frac{2\pi}{7} = \frac{1}{2}$$

2

END OF TEST

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

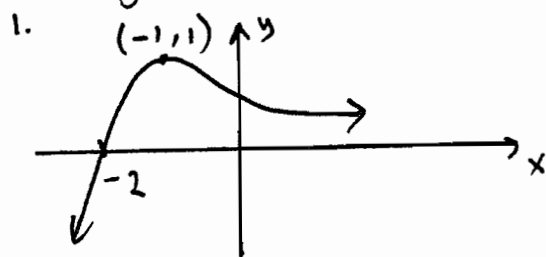
NOTE: $\ln x = \log_e x, \quad x > 0$

EXTENSION 2 : TASK II (2008)

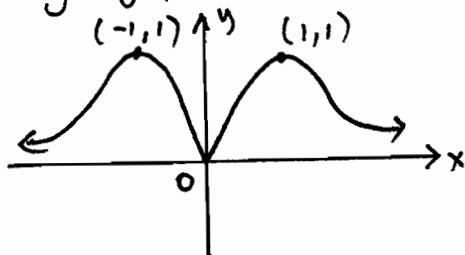
SOLUTIONS

Question 1

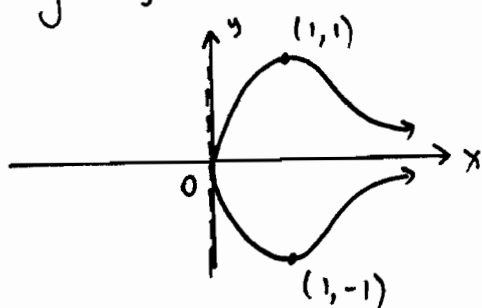
a) $y = f(x+2)$



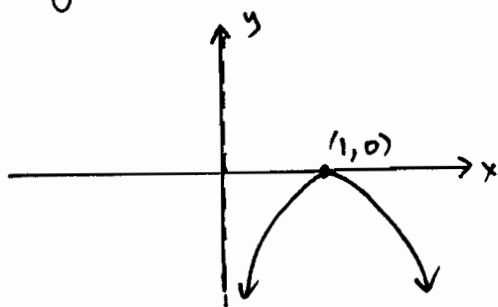
ii. $y = f|x|$



iii. $y^2 = f(x)$



iv. $y = \ln f(x)$



b) i. $\int_1^3 x \ln x \, dx$

$$= \ln x \cdot \frac{1}{2} x^2 - \int \frac{1}{2} x^2 \cdot \frac{1}{x} \, dx$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2} \int x \, dx$$

$$= \left[\frac{x^2 \ln x}{2} - \frac{x^2}{2} \right]_1^3$$

ii. $\int_0^\pi \sin x (1 - \cos^2 x) \, dx$

$$= \int_0^\pi \sin x - \sin x (\cos x)^2 \, dx$$

$$= \left[-\cos x + \frac{1}{3} \cos^3 x \right]_0^\pi$$

$$= -\cos \pi + \frac{1}{3} (\cos \pi)^3 - \left[-\cos 0 + \frac{1}{3} (\cos 0)^3 \right]$$

$$= 1 + \frac{1}{3} (-1)^3 - \left[-1 + \frac{1}{3} \right]$$

$$= 4/3$$

c) $\int \frac{dx}{\sqrt{5 + 4x - x^2}}$

$$= \int \frac{dx}{\sqrt{9 - (x-2)^2}}$$

$$= \sin^{-1} \left(\frac{x-2}{3} \right) + C$$

d) $\int_0^1 \frac{2}{\sqrt{x^2 + 1}} \, dx$

Standard $\int$ sheet

$$= 2 \ln \left[x + \sqrt{x^2 + 1} \right]_0^1$$

$$= 2 \left[\ln(1 + \sqrt{2}) - 0 \right]$$

$$= 2 \ln(1 + \sqrt{2})$$

Question 2

a)

$$x-2 = (Ax+B)(x+1) + C(x^2+4)$$

$$\text{let } x = -1$$

$$-3 = 5C$$

$$C = -3/5$$

$$\text{let } x = 0$$

$$B = 2/5$$

$$\text{let } x = 1$$

$$-1 = (A + 2/5)(2) + -3/5(5)$$

$$A = 3/5$$

$$\text{ii. } \int \frac{3x+2}{5(x^2+4)} - \frac{3}{5(x+1)} dx$$

$$= \int \frac{3x}{5(x^2+4)} + \frac{2}{5(x^2+4)} - \frac{3}{5(x+1)} dx$$

$$= \frac{3}{10} \ln(x^2+4) + \frac{1}{5} \tan^{-1} \frac{x}{2} - \frac{3}{5} \ln(x+1) + C$$

$$\text{b) } p(x) = x^5 + 2x^2 + mx + n$$

$$p'(x) = 5x^4 + 4x + m$$

$$p'(-1) = 0$$

$$5 - 4 + m = 0$$

$$m = -1$$

$$p(-1) = 0$$

$$-1 + 2 - m + n = 0$$

$$n = -2$$

$$\text{c) } 4x^3 + 8x^2 - 1 = 0$$

$$\text{let } x = \sqrt{x}$$

$$4\sqrt{x}^3 + 8\sqrt{x}^2 - 1 = 0$$

$$4x\sqrt{x} + 8x - 1 = 0$$

$$4x\sqrt{x} = 1 - 8x$$

$$16x^2 \cdot x = 1 - 16x + 64x^2$$

$$16x^3 - 64x^2 + 16x - 1 = 0$$

$$\text{ii. } \frac{1}{x^2} + \frac{1}{\beta^2} + \frac{1}{x^2}$$

$$\text{let } x = 1/x$$

$$16\left(\frac{1}{x}\right)^3 - 64\left(\frac{1}{x}\right)^2 + 16\left(\frac{1}{x}\right) - 1 = 0$$

$$16 - 64x + 16x^2 - x^3 = 0$$

$$x^3 - 16x^2 + 64x - 16 = 0$$

$$\therefore \frac{1}{x^2} + \frac{1}{\beta^2} + \frac{1}{x^2} = \frac{-b}{a} = 16$$

$$\text{d) } \int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx \quad x = 2\sin\theta$$

$$\frac{dx}{d\theta} = 2\cos\theta$$

$$= \int_0^{\pi/6} \frac{4\sin^2\theta}{\sqrt{4-4\sin^2\theta}} \cdot 2\cos\theta d\theta \quad dx = 2\cos\theta d\theta$$

$$x=1 \quad \theta = \pi/6$$

$$x=0 \quad \theta = 0$$

$$= \int_0^{\pi/6} \frac{8\sin^2\theta \cos\theta d\theta}{2\cos\theta}$$

$$= 4 \int_0^{\pi/6} \sin^2\theta d\theta$$

$$2 \int 1 - \cos 2\theta d\theta$$

$$= 2 \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/6}$$

$$= 2 \left[\frac{\pi}{6} - \frac{\sqrt{3}}{4} - (0) \right]$$

$$= \frac{\pi}{3} - \frac{\sqrt{3}}{2}$$

Question 3.

a) $P(x) = ax^4 + bx^3 + dx + e$

$$P'(x) = 4ax^3 + 3bx^2 + d$$

$$P''(x) = 12ax^2 + 6bx = 0$$

$$x(12ax + 6b) = 0$$

$x=0$ is not Triple root as $P(0) \neq 0$

$$\therefore 12ax + 6b = 0$$

$$x = -\frac{6b}{12a}$$

$$x = -\frac{b}{2a}$$

$$\therefore P'\left(-\frac{b}{2a}\right) = 4a\left(-\frac{b}{2a}\right)^3 + 3b\left(-\frac{b}{2a}\right)^2 + d = 0$$

$$-\frac{4ab^3}{8a^3} + \frac{3b^3}{4a^2} + d = 0$$

$$-\frac{2b^3}{4a^2} + \frac{3b^3}{4a^2} + d = 0$$

$$\frac{b^3}{4a^2} + d = 0$$

i.e. $4a^2d + b^3 = 0$

b) $xy = c^2$

$$y = \frac{c^2}{x}$$

$$y' = -\frac{c^2}{x^2} \text{ at } x = cp$$

$$m_T = \frac{-c^2}{c^2 p^2} = -\frac{1}{p^2} \left(-\frac{1}{t^2}\right)$$

$$\therefore y - c/t = -\frac{1}{t^2}(x - ct)$$

$$t^2 y - tc = -x + ct$$

$$x + t^2 y = 2ct$$

at P $x + p^2 y = 2cp$

at Q $x + q^2 y = 2cq$

Finding R $(p-q)(p+q)y = 2c(p-q)$

$$x = 2cp - p^2 \left(\frac{2c}{p+q}\right) = \frac{2cpq}{p+q}$$

$$R = \left[\frac{2cpq}{p+q}, \frac{2c}{p+q} \right]$$

III. Gradient PS = Gradient QS

$$\frac{c/p - c\sqrt{2}}{cp - c\sqrt{2}} = \frac{c/q - c\sqrt{2}}{cq - c\sqrt{2}}$$

$$\frac{1/p - \sqrt{2}}{p - \sqrt{2}} = \frac{1/q - \sqrt{2}}{q - \sqrt{2}}$$

$$\frac{q}{p} - \frac{\sqrt{2}}{p} - \sqrt{2}q + 2 = \frac{p}{q} - \sqrt{2}p - \frac{\sqrt{2}}{q} + 2$$

$$q^2 - q\sqrt{2} - \sqrt{2}pq^2 = p^2 - p\sqrt{2} - \sqrt{2}p^2q$$

$$\sqrt{2}p(1+pq) - \sqrt{2}q(1+pq) = (p-q)(p+q)$$

$$\sqrt{2}(p-q)(1+pq) = (p-q)(p+q)$$

$$\therefore p+q = \sqrt{2}(1+pq)$$

IV. $\frac{p+q}{\sqrt{2}} - 1 = pq$

$$x = \frac{2cpq}{p+q}, \quad y = \frac{2c}{p+q}$$

$$x(p+q) = 2cpq, \quad (p+q) = \frac{2c}{y}$$

$$x \left[\frac{2c}{y} \right] = 2c \left[\frac{p+q}{\sqrt{2}} - 1 \right]$$

$$\frac{2xc}{y} = 2c \left[\frac{2c/y}{\sqrt{2}} - 1 \right]$$

$$xc = cy \left[\frac{2c}{\sqrt{2}y} - 1 \right]$$

$$x = y \left[\frac{\sqrt{2}c}{y} - 1 \right]$$

$$d) z^7 = 1$$

$$z = 1 \text{ is a sol}$$

$$z = 1 \text{ cis } 0$$

$$z = \left[1 \text{ cis } (0 + 2\pi k) \right]^{1/7}$$

$$z = \text{cis} \left(\frac{0 + 2\pi k}{7} \right)$$

$$k=0 \quad z = 1$$

$$k=1 \quad z = \text{cis } 2\pi/7$$

$$k=2 \quad z = \text{cis } 4\pi/7$$

$$k=3 \quad z = \text{cis } 6\pi/7$$

$$k=4 \quad z = \text{cis } 8\pi/7$$

$$k=5 \quad z = \text{cis } 10\pi/7$$

$$k=6 \quad z = \text{cis } 12\pi/7$$

ii) sum roots

$$\begin{aligned} & \text{cis } \frac{2\pi}{7} + \text{cis } \frac{4\pi}{7} + \text{cis } \frac{6\pi}{7} + \text{cis } \frac{8\pi}{7} + \text{cis } \frac{10\pi}{7} \\ & + \text{cis } \frac{12\pi}{7} = -1 \end{aligned}$$

$$2 \cos \frac{2\pi}{7} + 2 \cos \frac{4\pi}{7} + 2 \cos \frac{8\pi}{7} = -1$$

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2}$$

$$\cos \frac{2\pi}{7} - \cos \frac{3\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2}$$

$$-\cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} + \cos \frac{\pi}{7} = -\frac{1}{2}$$

$$\cos \frac{3\pi}{7} + \cos \frac{\pi}{7} - \cos \frac{2\pi}{7} = \frac{1}{2}$$